

Repo - [open-WFS](#) & [open-WFS-software](#)

Spat account (auth.ircam.fr):

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Live 12 ([ableton.com](#)):

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Terms:

- Multi-output device = same output through all devices
- Aggregate device = control different outputs of multiple devices as 1
- Wave field synthesis - synthesizes waveforms

Downloads

1. REAPER
2. Blackhole 64ch
3. Max/MSP 9
4. Spat

System Setup Steps:

1. 169.254.100.100
 - a. Netgear address to figure for AVB
2. Plug in Ethernet & then power cables of modules
 - a. To reset: reboot network switch → *plug speakers back in* → *restart computer*
3. (If using OSC): plug in other computer using Ethernet afterwards
4. Audio MIDI Setup > Window > Show Network Device Browser:
 - a. Check boxes for all modules (should show up as devices)
5. Audio Devices > Aggregate Device: check boxes for all modules
 - a. To order speakers: drag nodes using Reaper example file
 - b. Verify speakers are working by playing file: open-wfs examples > Reaper Broadcast 1-8
6. OpenWFS.maxpat file (openwfs-software > max)
 - a. Setup speaker layout (info below)
 - b. Input: BlackHole 64ch, output: Aggregate device
 - c. @sources: 8, @speakers: 256, @channels: 256
 - d. Click dump
 - e. *Ensure button is green/"Audio is ON"
7. (Search system) Sound > BlackHole 64ch
8. Open Reaper
 - a. Settings (opens Reaper Preferences) > Audio > Device > Audio Device: choose BlackHole 64ch
 - i. *to test with only REAPER (not through Spat): Audio Device: Aggregate Device
 - b. Open file to play

- i. Each channel through which sound is output to on Reaper corresponds to the sound source object (ie Output Channel 1 = Sound Object 1)
- c. Click play button
- d. Open spat5.viewer & drag around sound objects
 - i. Alternatively use OSC to control object position
 - ii. Make sure there is a wire from the port to the spat viewer

Troubleshooting:

- Software:
 - Full reset (info above)
 - Make sure Audio is ON & Reaper volume + Spat volume + system volume is up
 - May be quiet: all the way
 - Sound objects may be too far away
 - Make sure all input & output devices are correct
- Hardware:
 - Make sure all power cords are firmly in wall
 - Connector can get loose so green light won't turn on
 - Make sure no additional devices are connected to switch
 - Reboot the network switch

To edit speaker layout:

- Activate virtual environment:
- Make sure openwfs folder was copied into the egg
- Change config/constants file:
 - NOT the file in the open-wfs-software folder. You must edit the file in the egg (.../.egg/openwfs/constants.py)
- Generate speaker layout:
 - `python3 openwfs-software/bin/generate-spat-layout.py max/speaker-layout.txt`
- Reopen .maxpat file in Max
- Click dump button

To set up track with different individual sound objects:

- Drag each track into Reaper
- Create specified number of sound objects in Spat
- Click play button
- Problem: each track is stereo but sound objects can't take stereo: each needs to take only 1
 - By default wants to send both channels out: would go to channels 1 & 2
- Solution: for each track: click on Routing (diagonal green striped box near volume controls)
 - Bottom left corner: mono source
 - Uncheck master box
 - Add new hardware output: scroll to mono → output 1, output 2, etc.

Sending OSC Coordinates with Script:

- Must be on same network (make sure IP addresses & port (udpreceive) match):
 - Connect laptop to Ethernet switch using adapter + cable
- *OSC command: /source/[#]/xyz [x-coord] [y-coord] [z-coord]
- *must send floats for OSC
- From port on max patch, add line to the object location patch

Understanding Max:

- Must install Max ? (wavefield synthesis system)
 - Trying to run spatializer in Max For Live but uses input & output for Ableton when Ableton is open
 - Max for Live: integration for max that allows you to do most of things but can't run max for live & another max patch at same time on separate audio devices
- To be able to use Ableton (preferred method) & Spat: need to open Spat in MaxMSP
 - Max For Live is wrapped in Ableton: so changing Ableton changes Max For Live
 - Ableton not open: can set input & output device
 - Ableton open (play sound through Ableton and set inputs & outputs on Ableton): can't be different than ones on Max
 - *Won't be able to make changes in Max but will in Max For Live*
 - *Spat doesn't run properly in Max For Live*
- Everything based around origin (head) at 0, 0, 0: JP's transforms would be wrong if moved head
- From reaper to blackhole to spat to audio device
 - REAPER: output = BlackHole 64ch → MaxMSP: input = BlackHole 64ch, output = Aggregate Device
- AVB happening directly out of Spat

Module dimensions: height = 0.15m, length = 1.065m

Troubleshooting 8 speakers

- Use wireshark to examine packets going through the network
 - eth.type == 0x22f0: AVB audio stream packets
 - eth.type == 0x88f5: talker/listener registration
 - eth.type == 0x88f7: clock
- Looking for
 - Registration of all modules
 - Audio stream sent from mac, received by module
 - Syncing of all modules
 - Based on mac clock